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2027 CYCLE

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CUET UG

Chemistry

Chemical Kinetics

Official Syllabus Unit III · Class 12 · CUET UG 2027 Cycle Notes

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment

Sub-topic	Official syllabus pointer (NTA CUET(UG) Chemistry syllabus 2026, the current 2027 baseline, Unit III)	Skills the questions test
Rate and its measurement	Introduction to chemical kinetics, Rate of a chemical reaction (Average and instantaneous rate of reactions)	Writing rate expressions with stoichiometric divisors, converting between rates of different species, and reading average vs instantaneous rates off concentration-time data.
Factors and rate law	Factors influencing rate of reaction: concentration, temperature, catalyst, Rate law and rate constant, Order and molecularity of a reaction	Deducing the rate law from experimental data, stating units of k for each order, and contrasting order (experimental) with molecularity (theoretical).
Integrated equations and half-life	Integrated rate equations and half-life of a reaction (for zero and first order reactions only)	Using $[A] = [A]_0 - kt$ and $k = \left(\frac{2.303}{t}\right) \log\left(\frac{[A]_0}{[A]}\right)$, computing half-lives, and predicting how $t_{\frac{1}{2}}$ depends on concentration.
Pseudo first order	Pseudo first order reactions	Recognising ester hydrolysis and sugar inversion as second-order reactions that behave first-order because water sits in huge excess.
Arrhenius and collision theory	Arrhenius equation, activation energy, collision theory for a bimolecular elementary chemical reactions and its limitations	Applying $k = Ae^{-\frac{E_a}{RT}}$, using the two-temperature log form to find E_a, and explaining the collision criteria and where the theory breaks down.

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EXAM PATTERN

CUET UG pattern (NTA official information bulletin and syllabus, 2026 cycle — the current 2027 baseline): each subject paper carries **50 compulsory questions** in **60 minutes**, with +5 for every correct answer and −1 for every incorrect answer. The Chemistry paper maps strictly to NCERT Class 12, and Unit III is the most numerical unit of physical chemistry on the paper: rate laws, integrated equations and Arrhenius calculations.

Rate of a Reaction: How Fast Chemistry Happens

Some reactions finish in a flash — a firework, a precipitate forming the instant two solutions meet. Others crawl: iron rusting over years, or a candle burning steadily for hours. **Chemical kinetics** is the study of this "how fast" — the speeds of reactions, the factors that steer them, and the step-by-step paths (mechanisms) they take. Thermodynamics told us whether a reaction **can** happen; kinetics tells us whether it **will** happen in our lifetime. Diamond turning into graphite is thermodynamically welcome, yet the rate is so tiny that your ring survives — that gap between "allowed" and "actual" is the whole subject.

- **Rate = change in concentration per unit time:** for a reactant R disappearing,

$$\text{average rate} = -\frac{\Delta[R]}{\Delta t}$$

  The minus sign is a courtesy, not physics: reactant concentrations fall, and rates are reported as positive numbers.
- **Average rate:** measured over a chosen stretch of time — total change divided by the interval. It smooths over the fact that the speed keeps changing.
- **Instantaneous rate:** the rate at one exact moment — the slope of the tangent to the concentration-time curve at that point:

$$\text{instantaneous rate} = -\frac{d[R]}{dt}$$

- **Rates change as the reaction proceeds:** the tangent starts steep and flattens — reactions are fastest at the start (most reactant, most collisions) and slow down as reactants run out.
- **Units:** concentration per time —

$$\text{mol L}^{-1}\text{s}^{-1}$$

  — the same units for every order of reaction, a fact that catches many students off guard.

One reaction, several rates — the stoichiometric divider

In

$$N_2 + 3H_2 \rightarrow 2NH_3$$

, nitrogen disappears 3 times slower than hydrogen vanishes (per mole) and ammonia appears twice as fast as nitrogen goes. To report ONE rate for the reaction, divide each species' rate by its coefficient:

$$\text{rate} = -\frac{d[\text{N}_2]}{dt} = -\left(\frac{1}{3}\right)\frac{d[\text{H}_2]}{dt} = +\left(\frac{1}{2}\right)\frac{d[\text{NH}_3]}{dt}$$

WORKED EXAMPLE — CONVERTING BETWEEN SPECIES RATES

In the decomposition $2\text{N}_2\text{O}_5 \rightarrow 4\text{NO}_2 + \text{O}_2$, the

$$\text{N}_2\text{O}_5$$

concentration falls from 0.50 M to 0.44 M in 60 s. Find (a) the average rate of disappearance of

$$\text{N}_2\text{O}_5$$

, (b) the rate of appearance of

$$\text{NO}_2$$

, (c) the reaction rate.

Part (a).

$$\text{rate of disappearance} = \frac{0.50 - 0.44}{60} = \frac{0.06}{60} = 1.0 \times 10^{-3}\text{M s}^{-1}$$

Part (b).

$$\text{NO}_2$$

appears 4 times faster than

$$\text{N}_2\text{O}_5$$

disappears (coefficients $4 : 2 = 2 : 1$):

$$\frac{d[\text{NO}_2]}{dt} = 2 \times 1.0 \times 10^{-3} = 2.0 \times 10^{-3}\text{M s}^{-1}$$

Part (c). The reaction rate divides by the coefficient:

$$\text{rate} = \left(\frac{1}{2}\right)(1.0 \times 10^{-3}) = 5.0 \times 10^{-4}\text{M s}^{-1}$$

The habit to build: always glance at the coefficients before quoting any rate — exam options usually include the un-divided and mis-divided numbers as bait.

What Controls the Speed of a Reaction

Four levers move reaction rates, and the syllabus names three of them explicitly. Each lever works by changing either **how often** molecules collide, **how hard** they collide, or **how easy the energy hill** is to cross.

- **Concentration:** more reactant particles per litre means more collisions per second — and more chances for successful ones. Gases under higher pressure behave the same way (squeezed = crowded).
- **Temperature:** heat hands molecules more kinetic energy, so a bigger fraction can cross the activation hill. The classic rough rule: a 10°C rise **nearly doubles** the rate — the

Arrhenius equation (later in this chapter) explains why so small a temperature change does so much.

- **Catalyst:** a substance that speeds up a reaction **without being consumed**, by offering an alternative path with a lower activation energy. Two honest facts follow: it changes **neither ΔH nor** the equilibrium constant — it only helps equilibrium arrive faster; and anything that lowers the hill for the forward reaction lowers it equally for the reverse one.
- **Surface area (the unnamed fourth lever):** powdered solids react faster than lumps — more exposed particles, more collision sites. Wood shavings burn faster than a log; the chemistry is identical.

ANALOGY — THE SKI SLOPE AND THE TUNNEL

Picture the reaction as a skier crossing a mountain: reactant valley on one side, product valley on the other, and a pass between them whose height is the activation energy. **Temperature** sends skiers up with more energy — more of them can clear the same pass. **A catalyst** does not push anyone; it digs a lower tunnel through the ridge, so even average skiers can cross. The two valleys (reactant energy and product energy) never move — which is exactly why a catalyst changes the speed but not the destination, the energy released, or the equilibrium position.

Rate Law and Rate Constant: Let the Data Speak

Watch a reaction's rate respond to concentration and you can write its **rate law** — an equation of the form Rate $= k[A]^x[B]^y$. But here is the rule that separates students who understand kinetics from those who guess: the exponents x and y are found only from experiment. They are usually **not** the coefficients of the balanced equation.

- **Rate law (rate equation):** for $aA + bB \rightarrow$ products, the observed law is Rate $= k[A]^x[B]^y$ where x, y come from controlled measurements — vary $[A]$ with $[B]$ fixed, watch how the rate answers.
- **Order of reaction:** the sum $x + y$. It can be zero, a whole number, a fraction ($\frac{3}{2}$), even negative — whatever the data say.
- **Rate constant (k):** the proportionality constant — the rate when every concentration is 1 M. Its value depends on temperature (and on the catalyst) but **not** on concentrations.
- **The units of k change with order** — a favourite one-mark check:

Overall order	Units of k	Quick way to remember
Zero	mol L^{-1} s^{-1}	Rate does not use concentrations — k carries the rate's own units
First	s^{-1}	Rate/concentration: (mol L^{-1} s^{-1})/(mol L^{-1})
Second	L mol^{-1} s^{-1}	Divide the rate unit by two concentration powers

Overall order	Units of k	Quick way to remember
Third	$L^2\ mol^{-2}\ s^{-1}$	One more ($L\ mol^{-1}$) added per order step

Deducing a rate law from data — the comparison drill

WORKED EXAMPLE — FINDING X AND Y FROM A DATA TABLE

For the reaction $A + B \rightarrow$ products:

Experiment	Initial [A] (M)	Initial [B] (M)	Initial rate ($M\ s^{-1}$)
1	0.10	0.10	3.0×10^{-4}
2	0.20	0.10	1.2×10^{-3}
3	0.20	0.20	1.2×10^{-3}

Step 1 — how does [A] act? Compare experiments 1 and 2 ([B] fixed): $[A]$ doubles, and the rate goes from 3.0 to 12 ($\times$ 4). Since $2^x = 4$, $x = 2$ — second order in A.

Step 2 — how does [B] act? Compare experiments 2 and 3 ([A] fixed): $[B]$ doubles but the rate does not budge. So $y = 0$ — zero order in B.

Step 3 — write the law. Rate $= k[A]^2[B]^0 = k[A]^2$. Overall order = 2.

Step 4 — find k. Use experiment 1:

$$k = \frac{3.0 \times 10^{-4}}{(0.10)^2} = \frac{3.0 \times 10^{-4}}{0.01} = 3.0 \times 10^{-2} \text{L mol}^{-1}\text{s}^{-1}$$

Note how the units of k were **chosen** to match the second-order law — that is why they differ from order to order.

Order vs Molecularity: Two Numbers That Are Not Twins

Property	Order	Molecularity
What it is	Sum of exponents in the observed rate law	Number of molecules colliding in one elementary step
Where it comes from	Experiment only	Theory — read off the proposed mechanism's step
Possible values	0, whole, fractional, negative	Only whole numbers 1, 2, 3 — never zero or fractional
Applies to	The overall reaction, however complex	Single elementary steps only

Property	Order	Molecularity
Can change?	Same for given conditions; can be "wrong-footed" by mechanisms	Fixed by the step you wrote

- **Why complex reactions hide their order:** a multi-step reaction is controlled by its **slowest step** (the rate-determining step). The overall rate law mirrors that step — plus any fast pre-equilibria feeding it — so the balanced equation's coefficients are irrelevant.
- **Elementary steps behave directly:** only for a single-step (elementary) reaction does molecularity equal the order. That coincidence is the exception, not the rule.
- **Zero-order reactions exist:** a reaction can run at constant speed independent of concentration — for instance when a saturated reactant surface or enzyme active site is completely occupied.

Integrated Rate Equations: Zero and First Order

The rate law tells you the speed at an instant; the **integrated rate equation** tells you the concentration at **any** time — the difference between a speedometer and an odometer. The syllabus restricts the maths to zero and first order, which is exactly what you need.

Zero order: the conveyor belt

Rate is constant, independent of concentration. Integrating Rate $= k$ gives:

$$[A] = [A]_0 - kt$$

Concentration falls in a **straight line** against time (slope $-k$). The half-life follows from setting $[A] = \frac{[A]_0}{2}$:

$$t_{\frac{1}{2}} = \frac{[A]_0}{2k}$$

- **Read the half-life formula like a sentence:** the more you start with, the longer the first half takes — $t_{\frac{1}{2}}$ is **proportional** to $[A]_0$. Each successive half-life is half the previous one, because the concentration keeping the belt running keeps dropping.
- **Where zero order really happens:** enzyme-saturated biochemistry, and metal-surface catalysis where the surface is fully coated (the classic

  $$\mathrm{NH_3}$$

  on hot platinum example).

First order: the radioactive standard

Rate $= k[A]$. Integrating (with natural logs, then converting to base 10) gives the exam workhorse:

$$k = \left(\frac{2.303}{t}\right) \log\left(\frac{[A]_0}{[A]}\right)$$

- **Half-life:** set $[A] = \frac{[A]_0}{2}$:

$$t_{\frac{1}{2}} = \frac{0.693}{k}$$

Look hard at what is **missing**: the starting concentration. A first-order half-life is **the same** whether you begin with 10 M or 0.001 M. That independence is the signature of first-order behaviour — and the reason radioactive dating works so cleanly.

- **Each half-life halves whatever remains:** 100% → 50% → 25% → 12.5%. To reach 75% completion you must wait **two** half-lives (only 25% remains).
- **Straight-line test:** plot $\log[A]$ against t — first order gives a straight line with slope $-\frac{k}{2.303}$. Plot $[A]$ against t: a zero-order reaction is the straight line; first order curves downward.

WORKED EXAMPLE — FIRST ORDER, END TO END

A first-order reaction has a rate constant $k = 5.5 \times 10^{-2}\text{s}^{-1}$. Find (a) its half-life, (b) the time for the concentration to fall to one-tenth of its initial value.

Part (a).

$$t_{\frac{1}{2}} = \frac{0.693}{k} = \frac{0.693}{5.5 \times 10^{-2}} \approx 12.6 \text{ s}$$

Part (b). One-tenth remaining means $\frac{[A]_0}{[A]} = 10$:

$$t = \left(\frac{2.303}{k}\right) \log 10 = \frac{2.303}{5.5 \times 10^{-2}} \approx 41.9 \text{ s}$$

Sanity check: falling to one-tenth needs more than three half-lives' worth of progress? Three half-lives leave $\frac{1}{8} = 0.125$ — so one-tenth comes slightly **after** three half-lives: $3 \times 12.6 = 37.8 < 41.9$ ✓. The log method and the half-life ladder agree.

Pseudo First Order Reactions: When One Molecule Hides

Here is a puzzle the syllabus loves: the hydrolysis of an ester,

$$\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{ O} \rightarrow \text{CH}_3\text{ COOH} + \text{C}_2\text{H}_5\text{ OH}$$

uses **two** reactants, yet experiments show the rate depends on the ester concentration **alone**. A second-order reaction pretending to be first-order is called a **pseudo first order** reaction — "pseudo" means fake.

- **The trick is excess:** water is the solvent — about 55.5 M in every litre. As the ester (say 0.01 M) reacts, the water concentration moves from 55.50 to 55.49 M — a change too tiny to matter. Effectively constant, it gets folded into k.
- **The rate law in practice:** Rate $= k'[\text{ester}]$ with $k' = k[\text{H}_2\text{ O}]$ — the hidden reactant rides inside the effective constant.
- **Sugar inversion, the second classic:** sucrose hydrolyses to glucose + fructose with acid catalyst; again water is in vast excess, so the disappearance of sucrose follows first-order kinetics (polarimetry tracks it, thanks to the optical rotation change).
- **How to spot it in a question:** a two-molecule equation whose measured kinetics come out cleanly first-order — answer: one reactant is secretly the solvent.

The Arrhenius Equation and Activation Energy

Why does a 10°C warming roughly double a reaction's speed? Svante Arrhenius answered with one of chemistry's most durable equations:

THE ARRHENIUS EQUATION

$$k = Ae^{-\frac{E_a}{RT}}$$

Letters: A = the **frequency factor** (how often molecules collide with the right geometry), E_a = **activation energy** — the energy hill of Figure 1, R = gas constant (8.314 J K^{-1} mol^{-1}), T = absolute temperature in kelvin.

- **The exponential is the star:** the fraction of molecules carrying energy at least E_a behaves like $e^{-\frac{E_a}{RT}}$ — a tiny-looking exponent change swings k enormously. That is the engine behind the 10°C rule.
- **Taking logs** gives the straight-line form for graph work:

  $$\log k = \log A - \frac{E_a}{2.303RT}$$

  Plot $\log k$ against $\frac{1}{T}$: a straight line with **slope** $= -\frac{E_a}{2.303R}$ and intercept $\log A$. Read E_a straight off the slope.
- **Two temperatures, no graphs:** subtract the log form at T_1 from T_2:

  $$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$$

  This is the single most-quoted kinetics formula in exams.
- **What E_a means:** it is **not** heat released or absorbed — it is the barrier height between the reactant plateau and the transition state. Reactions with low E_a run fast at room temperature; high-E_a reactions need flame or enzyme.

WORKED EXAMPLE — ACTIVATION ENERGY FROM TWO RATE CONSTANTS

The rate constant of a reaction doubles when the temperature rises from 300 K to 310 K. Find the activation energy. ($R = 8.314$ J K^{-1} mol^{-1}.)

Step 1 — state the shortcut with $\frac{k_2}{k_1} = 2$.

$$\log 2 = \frac{E_a}{2.303 \times 8.314}\left(\frac{310 - 300}{300 \times 310}\right)$$

Step 2 — the temperature side.

$$\frac{310 - 300}{300 \times 310} = \frac{10}{93000} = 1.075 \times 10^{-4} \mathrm{K}^{-1}$$

Step 3 — substitute and solve.

$$\log 2 = 0.3010$$

, so

$$0.3010 = E_a \times 1.075 \times \frac{10^{-4}}{19.15} \rightarrow E_a = \frac{0.3010 \times 19.15}{1.075 \times 10^{-4}}$$

Step 4 — arithmetic.

$$E_a = \frac{5.764}{1.075 \times 10^{-4}} \approx 5.36 \times 10^4 \text{J mol}^{-1} \approx 53.6 \text{kJ mol}^{-1}$$

Read the lesson: a mere 10 K of warming doubling the rate is the signature of a barrier around 50 kJ mol^{-1} — typical for many everyday reactions. Barriers near 20 kJ give explosive temperature sensitivity; near 200 kJ, reactions barely move without a furnace.

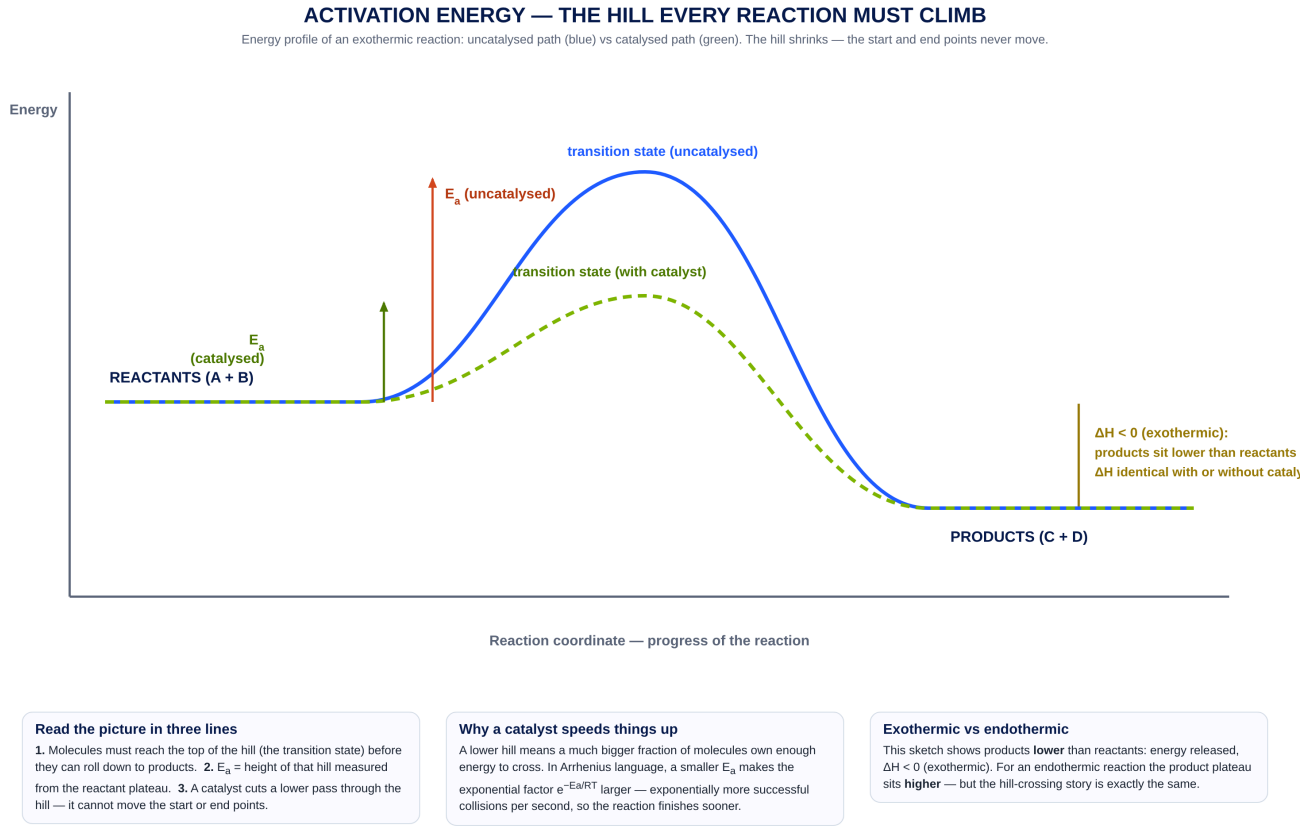


Figure 1: Activation energy on an energy profile: the uncatalysed hill, the lower catalysed pass, and the unchanged reactant/product levels that fix ΔH.

Collision Theory: Why Reactions Need Luck

For a bimolecular elementary step $A + B \rightarrow$ products, **collision theory** builds rate from the bottom up: molecules must collide, and only special collisions count.

- **Criterion 1 — the energy test:** the colliding pair must carry total kinetic energy at least equal to the **threshold energy** (E_a plus a small allowance). Molecules below the line simply bounce off.
- **Criterion 2 — the orientation test:** energy alone is not enough; the molecules must meet in the **right geometry** so old bonds can stretch and new ones form. The fraction of properly aligned collisions is the **probability or steric factor** P — often small (sometimes 10^{-5}), which is why A is much smaller than the raw collision frequency.

- **Putting it together:** Rate = collision frequency × energy fraction × orientation fraction. The energy fraction is exactly the Arrhenius exponential — collision theory **derives** the equation you already use.
- **The limitations (the syllabus asks for them):** it treats molecules as hard, featureless spheres — so it cannot explain why some structurally complex collisions with ample energy still fail (bond-by-bond rearrangements matter); it has no clean way to predict P from first principles; and it says little about multi-step mechanisms. Transition-state theory fixes these — but that is a university story.

Key Reactions & Rules: One-Stop Revision

RATES, ORDERS, HALF-LIVES, ARRHENIUS

Rule / formula	Symbols in plain words	When to reach for it
$\text{rate} = -\frac{\Delta[R]}{\Delta t}$	minus sign keeps the rate positive	Average rate from concentration change
$\text{rate} = -\left(\frac{1}{\nu}\right)\frac{d[R]}{dt}$	divide each species by its coefficient	Any multi-species rate statement
$\text{Rate} = k[A]^x[B]^y$	x, y from experiment only	Writing/interpreting rate laws
k units by order	0: mol L^{-1} s^{-1}; 1: s^{-1}; 2: L mol^{-1} s^{-1}	Identifying hidden order from units
$[A] = [A]_0 - kt$	zero order; straight-line [A] vs t, slope −k	Constant-rate (surface/enzyme) reactions
$t_{\frac{1}{2}} = \frac{[A]_0}{2k}$	zero order: halves shrink each time	Zero-order half-life questions
$k = \left(\frac{2.303}{t}\right)\log\left(\frac{[A]_0}{[A]}\right)$	first order; log [A] vs t straight, slope −k/2.303	Every first-order concentration question
$t_{\frac{1}{2}} = \frac{0.693}{k}$	no $[A]_0$ anywhere — the signature	Half-life independence checks
$\text{Rate} = k'[\text{ester}]$	$k' = k[\text{H}_2\text{O}]$ with water in excess	Pseudo first order (ester, sugar)
$k = Ae^{-\frac{E_a}{RT}}$	A: collision frequency factor; E_a: the hill	Temperature dependence of k
$\log k = \log A - \frac{E_a}{2.303RT}$	slope of log k vs 1/T $= -\frac{E_a}{2.303R}$	Graph-based E_a questions

Rule / formula	Symbols in plain words	When to reach for it
$\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2-T_1}{T_1T_2}\right)$	no graphs needed	Two-temperature E_a calculations
Rate = frequency × energy × orientation	steric factor P for geometry	Collision-theory explanations

Exceptions & Traps

THE TRAPS THIS CHAPTER SETS — READ BEFORE THE MOCK

The rule you assume	The exception / twist	The trap in the question
Exponents match the equation's coefficients	Order is experimental — often different	Rate $= k[NO_2]^2$ for a $2NO_2$-type equation is NOT automatic; check the data
Half-life always shortens as reactant runs out	First-order $t_{\frac{1}{2}}$ is constant from start to finish	The "half-life independent of concentration" check — first order only
k's units are always mol L^{-1} s^{-1}	Only zero order keeps the rate's units	Units given as s^{-1} or L mol^{-1} s^{-1} quietly reveal the order
A catalyst changes the equilibrium	It changes only the speed — ΔH, K and the position all stay put	"Catalyst increases yield" — false; it shortens the wait, not the destination
Temperature changes only the speed, not k	k itself is temperature-dependent (Arrhenius)	Rate-constant comparisons between temperatures need the two-point log equation
Two-reactant equation means second-order rate law	Water in huge excess hides — pseudo first order	Ester hydrolysis and sugar inversion: rate depends on the ester alone
75% completion takes double the half-life	Each half-life removes half of what remains	100 → 50 → 25: reaching 75% needs TWO half-lives
Molecularity and order are interchangeable	Molecularity is per-step, whole-number, never zero; order is overall, any number	A "molecularity = 1.5" option is instant bait

The Whole Chapter in One Picture

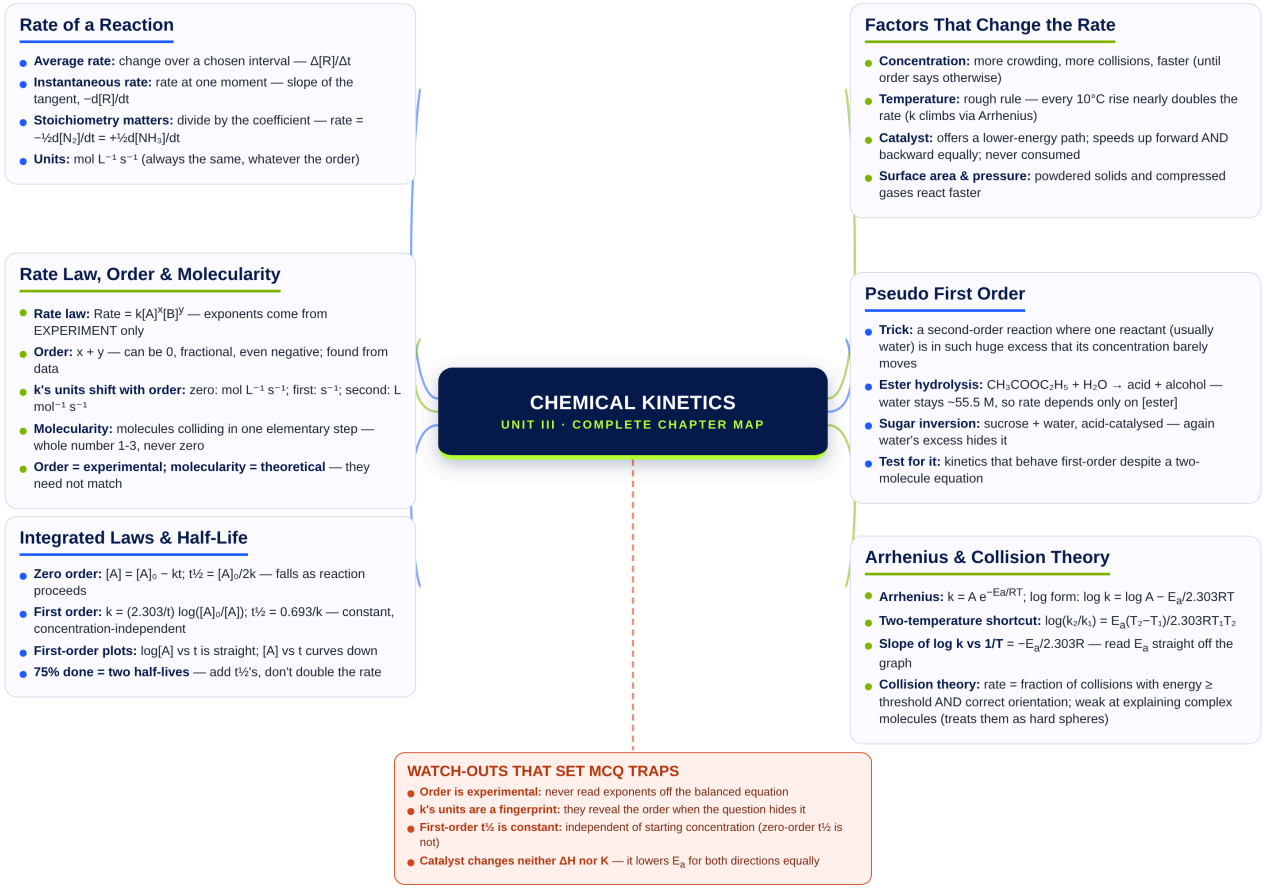


Figure 2: Chapter mindmap: rate and its types, the four factors, rate law and order vs molecularity, integrated zero/first-order laws with half-lives, pseudo first order, and Arrhenius with collision theory — with the traps that catch most students.

Practice Questions (Exam Pattern)

Ten questions in the CUET UG style. Attempt all of them — the paper gives +5 for each correct answer and takes nothing away for a wrong one.

Q1. The rate law of a reaction must be determined:

(A) from the balanced chemical equation
(B) only by experiment
(C) from the molecularity of the reaction
(D) from the charges of the ions

Q2. For a reaction A → products that is second order in A, the units of the rate constant k are:

(A) s^{-1}
(B) mol L^{-1} s^{-1}
(C) L mol^{-1} s^{-1}
(D) L^2 mol^{-2} s^{-1}

Q3. The half-life of a first-order reaction is 20 minutes. The fraction of reactant left after 60 minutes is:

(A) one-half
(B) one-quarter
(C) one-eighth
(D) one-sixteenth

Q4. For a zero-order reaction, a plot of concentration against time gives:

(A) a straight line with positive slope
(B) a straight line with slope $-k$
(C) a curve that levels off
(D) an exponential decay

Q5. In the reaction $2N_2O_5 \rightarrow 4NO_2 + O_2$, if $-d[N_2O_5]/dt = 0.02$ M s^{-1}, then $d[NO_2]/dt$ equals:

(A) 0.01 M s^{-1}
(B) 0.02 M s^{-1}
(C) 0.04 M s^{-1}
(D) 0.08 M s^{-1}

Q6. Hydrolysis of an ester in water follows first-order kinetics although two reactants react. This is because:

(A) the ester is a catalyst
(B) water is in such large excess that its concentration stays nearly constant
(C) the reaction is actually zero order
(D) acid produced changes the mechanism

Q7. A catalyst increases the rate of a reaction by:

(A) increasing the activation energy
(B) decreasing the activation energy
(C) making ΔH more negative
(D) shifting the equilibrium towards products

Q8. The rate constant of a reaction becomes 4 times when temperature rises from 300 K to 320 K. If $\log 4 \approx 0.60$, the activation energy is about ($R = 8.314$ J K^{-1} mol^{-1}):

(A) about 27 kJ mol^{-1}
(B) about 42 kJ mol^{-1}
(C) about 55 kJ mol^{-1}
(D) about 83 kJ mol^{-1}

Q9. According to collision theory, a successful collision requires:

(A) energy greater than the threshold and proper orientation
(B) only very high collision frequency
(C) identical molecular masses
(D) the presence of a catalyst

Q10. The molecularity of an elementary reaction can be:

(A) zero or fractional
(B) any positive fraction
(C) only 1, 2 or 3
(D) negative for reverse steps

Detailed Solutions

Sol. Q1. The balanced equation tells you the stoichiometry, never the speed law. Mechanisms — which experiments probe — decide the exponents. Classic example: $2N_2O_5 \rightarrow 4NO_2 + O_2$ happens to be first order, not second.

Answer: (B)

Sol. Q2. Second order means rate carries two concentration powers: units of k = (mol L^{-1} s^{-1}) ÷ (mol $L^{-1})^2$ = **L mol^{-1} s^{-1}**. The units-of-k table is the fastest fingerprint in this chapter.

Answer: (C)

Sol. Q3. 60 minutes = three half-lives. Each one halves the stock: 1 → 1/2 → 1/4 → 1/8. Constant first-order half-life makes this ladder exact — no calculus needed.

Answer: (C)

Sol. Q4. Zero order: $[A] = [A]_0 - kt$ — a straight line with slope $-k$ (concentration falls steadily even as it thins). The exponential decay shape belongs to first order, and the levelling-off curve is just an ordinary slowing reaction.

Answer: (B)

Sol. Q5. NO_2 appears with coefficient 4 while N_2O_5 disappears with coefficient 2 — twice as fast: $0.02 \times 2 = 0.04$ M s^{-1}. Always multiply by the coefficient ratio (4/2).

Answer: (C)

Sol. Q6. Water is the solvent at about 55.5 M; the ester sits around 0.01 M. Reacting away the ester shifts water by an invisible fraction, so water's concentration folds into k — a pseudo first-order reaction.

Answer: (B)

Sol. Q7. A catalyst offers a lower-energy route — a smaller E_a — so more collisions clear the barrier. It changes neither ΔH nor the equilibrium constant; both directions speed up equally.

Answer: (B)

Sol. Q8. Two-temperature formula: $\log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left(\frac{T_2 - T_1}{T_1 T_2}\right)$. So $0.60 = E_a \frac{20}{2.303 \times 8.314 \times 300 \times 320}$. Denominator: $2.303 \times 8.314 = 19.15$; $19.15 \times 96000 \approx 1.84 \times 10^6$. Then $E_a = 0.60 \times 1.84 \times \frac{10^6}{20} \approx 5.5 \times 10^4$ J mol^{-1} ≈ 55 kJ mol^{-1}.

Answer: (C)

Sol. Q9. Two gates must both open: enough energy (at least the threshold, i.e. E_a) **and** the right geometry for bonds to swap. Collision frequency alone counts every bump — including the millions that do nothing.

Answer: (A)

Sol. Q10. Molecularity counts molecules in one elementary step — you cannot have zero molecules collide, or 1.5 of one. Whole numbers 1 (unimolecular), 2, and rarely 3 only.

Answer: (C)

Wrong-Answer Clinic — the one-line corrections

Dropped marks above? Find your slip here and rewrite the corrected rule in your own words.

If you answered...	Your slip was...	The corrected rule, in one line
Q1 option A	Reading the equation as a law	Coefficients feed the **stoichiometric divider**, never the rate-law exponents — the data decide
Q3 option B	Counting half-lives as two	60 min ÷ 20 min = three halvings: 1/2 → 1/4 → 1/8
Q5 option B	Forgetting the coefficient ratio	Rate of a product = rate of reactant × (its coefficient ÷ reactant's)
Q7 option D	Confusing speed with extent	A catalyst gets you to equilibrium faster; the equilibrium position is untouched
Q8 option B	Dropping a factor in the denominator	$2.303RT_1T_2$ with T in kelvin — then divide by $(T_2 - T_1)$, not multiply
Q2 option A or B	Reusing the rate's units for k	k's units change with order because the law divides rate by concentration powers — second order always gives L mol^{-1} s^{-1}
Q10 option A or B	Imagining fractional molecularity	Molecularity counts whole molecules in one elementary step — 1, 2 or 3, and never zero; order is the number that can be fractional

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